INCH-POUND

MS25395J
w/AMENDMENT 1
26 March 2004
SUPERSEDING
MS25395H
5 Jun 1987

# RELAYS, ELECTROMAGNETIC, 5 AMPERES, 2 PDT, TYPE I, HERMETICALLY SEALED

INACTIVE FOR NEW DESIGN AFTER 15 NOVEMBER 2002.
NO SUPERSEDING SPECIFICATION.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

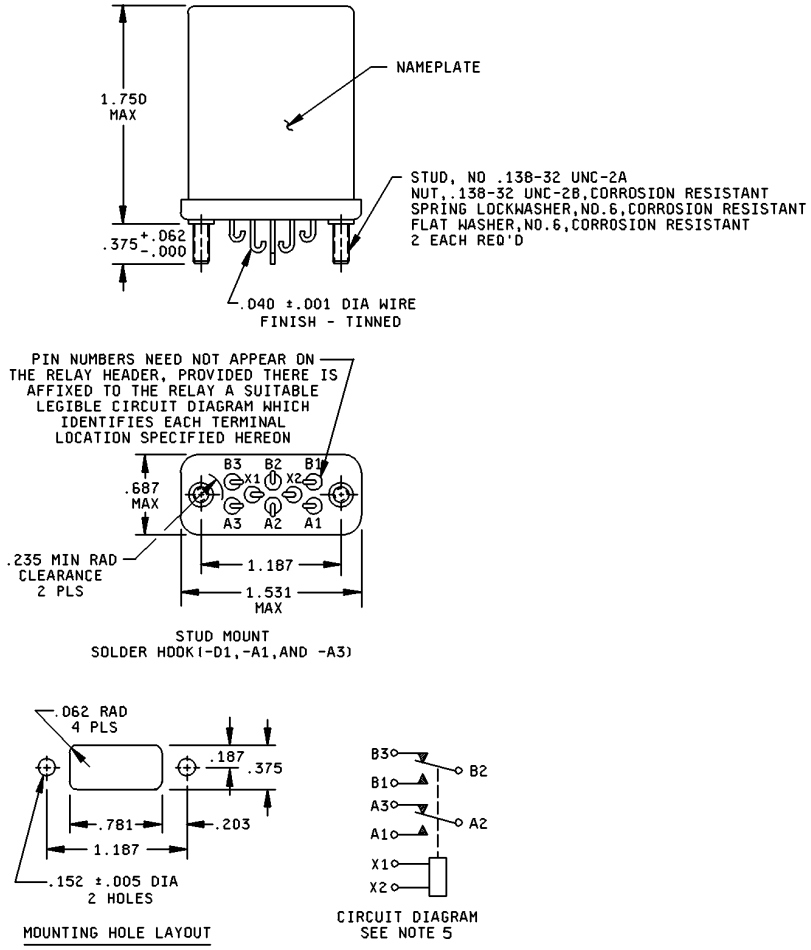


FIGURE 1. Dimensions and configurations.



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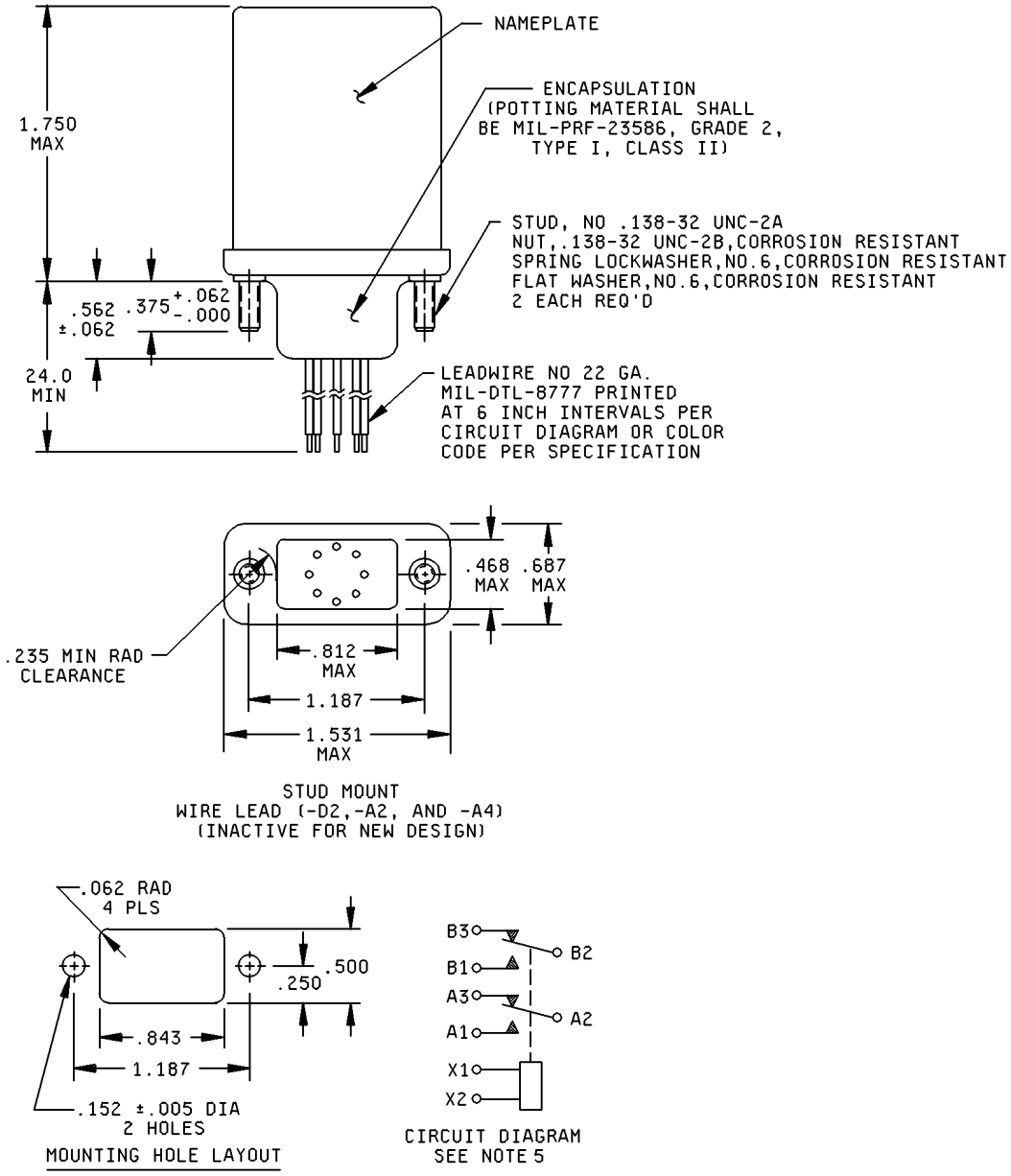


FIGURE 1. Dimensions and configurations - Continued.



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| Inches | mm | Inches | mm |
|---|---|---|---|
| .001 | 0.03 | .468 | 11.89 |
| .005 | 0.13 | .500 | 12.70 |
| .040 | 1.02 | .562 | 14.27 |
| .062 | 1.57 | .687 | 17.45 |
| .152 | 3.86 | .781 | 19.84 |
| .172 | 4.37 | .812 | 20.62 |
| .187 | 4.75 | .843 | 21.41 |
| .203 | 5.16 | 1.187 | 30.15 |
| .235 | 5.97 | 1.531 | 38.89 |
| .250 | 6.35 | 1.750 | 44.45 |
| .375 | 9.53 | 24.0 | 610. |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.
5. The use of diodes on ac relays is optional. Actual application must be shown on label.

FIGURE 1. Dimensions and configurations - Continued.

TABLE I. Dash numbers and characteristics.

| Dash number MS25395- | Type | Coil | Terminal type | Mounting or mating socket | Max weight in pounds |
|---|---|---|---|---|---|
| D1 | I | dc | Solder hook | Stud | 0.15 |
| D2 1/ | I | dc | Potted lead | Stud | 0.26 |
| A1 | I | ac | Solder hook | Stud | 0.17 |
| A2 1/ | I | ac | Potted lead | Stud | 0.28 |
| A3 | I | ac | Solder hook | Stud | 0.17 |
| A4 1/ | I | ac | Potted lead | Stud | 0.25 |

1/ CAUTION: The use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.



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TABLE II. Operating characteristics.

| PIN MS25395- | | Coil data | | | | | | | | | | | Time - (milliseconds maximum) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Coil | Nominal | | | Max | | Max pick-up voltage | | | Drop out volt-age 2/ | Hold volt-age 2/ | Oper -ate 3/ | Rel-ease 4/ | Contact bounce | | | |
| | | | | | | | | | | | | | | | Main | | Aux | |
| | | | Volts 1/ | Freq Hz | Res Ω ±10% | Volts | Amp | Nor-mal 2/ | High temp test | Cont cur-rent test | | | | | NO | NC | NO | NC |
| D1 | D2 | X1,X2 | 28 | dc | 248 | 29 | 0.15 | 18 | 19.8 | 22.5 | 1.5 | 7.0 | 20 | 20 | 2 | 2 | --- | --- |
| A1 | A2 | X1,X2 | 115 | 400 | N/A | 122 | 0.06 | 90 | 95 | 103 | 5.0 | 35 | 25 | 50 | 2 | 2 | --- | --- |
| A3 | A4 | X1,X2 | 115 | 50/ 60 | N/A | 122 | 0.07 | 90 | 95 | 103 | 5.0 | 35 | 25 | 50 | 2 | 2 | --- | --- |

1/ CAUTION: The use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.

TABLE III. Rated contact load (amperes per pole) (case grounded).

| Type of load | Life operating cycles x $10^3$ | 28 V dc | | | | 115 V ac, 1 phase | | | | 115/200 V ac, 3 phase 1/ | | | | See appro-priate notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | |
| Resistive | 100 | 5 | 5 | | | 5 | 4 | | | | | | | |
| Inductive | 100 | | | | | | | | | | | | | |
| Inductive | 20 | 3 | 3 | | | 3 | 2 | | | | | | | |
| Motor | 100 | 1.5 | 1.5 | | | 1.5 | 1 | | | | | | | |
| Lamp | 100 | 0.8 | 0.8 | | | 0.8 | 0.6 | | | | | | | |
| Transfer load | | | | | | | | | | | | | | 2/ |
| Mechanical life reduced current | 400 | 1.25 | 1.25 | | | 1.25 | 1 | | | | | | | |
| Mixed loads | Applicable in accordance with specification | | | | | | | | | | | | | |

1/ Absence of value indicates relay is not rated for 3-phase applications.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.



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Environmental characteristics.

| | |
|---|---|
| Temperature range | -70°C to +125°C |
| Max altitude rating | 80,000 ft |
| Shock G-level | 50 g's |
| Duration | 11 ms |
| Max duration contact opening | 10 μs |
| Vibration – sinusoidal | |
| G-level | 10 G |
| Frequency range | 5 - 1,500 Hz |
| Acceleration | 15 g's |

Electrical characteristics.

| | |
|---|---|
| Insulation resistance, initial | 100 megohms. |
| After life or environmental tests | 50 megohms. |

Dielectric strength (sea level).

| | Initial | After life tests |
|---|---|---|
| Coil to case | 1,000 V rms | 1,000 V rms |
| Aux contacts | N/A | N/A |
| All other points | 1,500 V rms | 1,125 V rms |

Dielectric strength (altitude):

| | 80,000 ft | -D2 -A2<br>80,000 ft |
|---|---|---|
| Coil to case | 250 V rms | 1,000 V rms |
| Aux contacts | N/A | N/A |
| All other points | 250 V rms | 250 V rms |

| | |
|---|---|
| Max contact drop initial: | 0.150 volt. |
| After life test: | 0.175 volt. |
| Overload current: | 20 amperes |
| Rupture current | 25 amperes |
| Duty rating: | Continuous. |
| RFI specification: | MIL-STD-461. |

(Applicable to coil circuits of ac operated relays).



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Conformance inspection.

Qualification by similarity: See MIL-PRF-6106.

Group B and C inspections may be suspended at the discretion of the qualifying activity.

Referenced documents. In addition to MIL-PRF-6106, this specification sheet references the following documents:

MIL-DTL-8777 MIL-STD-461
MIL-PRF-23586

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Navy - AS
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-1244)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at www.dodssp.daps.mil.



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